

Assignment 3

This project consists of 3 parts and 4 deliverables. Please submit your solutions by Friday, **26th of January 2026** at 11:59 am (noon). We expect you to work on the assignments in groups of three. All three students should contribute equally, work on all deliverables, and be able to present the report. For each assignment, deliver a report (.pdf) with the related Python code (.zip), covering all of the respective parts. Please submit the report and related code via Stud.IP, named according to assignment number and the names of each group member, e.g., *03_lastname_lastname_lastname.{pdf,zip}*.

We expect that each of you spends at most **23** hours on this assignment per group member. If you exceed this time limit without reaching a solution, submit your partial analyses and provide a detailed breakdown of the problems you encountered and how your time was distributed. In case of technical problems, we expect you to consult with our TA

- Surendhar Muthukumar (surendhar.m@stud.uni-goettingen.de)

before the deadline. Surendhar offers a Q&A session on Thursdays, from 11 am to 12:30 pm (Goldschmidtstr. 1, room 2.123), for technical support, understanding the tasks and lecture contents, and questions related to programming. For general organizational, assignment and lecture-related questions, contact the tutor

- Jonas Mayer Martins (jonas.mayermartins@uni-goettingen.de).

Part A: Language models in practice

(65 %)

Deliverable 1 – Task Analysis

Select a task from the **GLUE benchmark** for English and report relevant task statistics. Describe the task setup using examples from the dataset. Manually analyze the instances and select ten instances that you expect to be easy to solve and ten instances that you expect to be difficult.

Reflect on the annotation setup:

- What are the characteristics of the annotator sample?

- How detailed are the annotation guidelines? Are the scales anchored with examples?
- Is agreement between annotators reported? Are the raw annotations available?
- How were conflicting annotations merged?
- What is your opinion on the quality of the dataset (both the selection of instances and the annotations)?
- Explain your categorization of "easy" and "difficult" instances.

Deliverable 2 – Fine-tuning

Finetune GPT-2 on the GLUE task and compare the result to fine-tuning a randomly initialized GPT-2 model (no pre-training). Try at least two finetuning variants and compare the performance. Analyze the performance on the instance level for your 20 selected instances and reflect on the results.

Notebooks: *LaD3..._partial_finetuning.ipynb*, *LaD3..._LoRA.ipynb*, *LaD3..._instructions.ipynb*, and *LaD3..._DPO.ipynb*

Before adapting a finetuning setup to GLUE, start by working through the notebook on partial fine-tuning and LoRA, later move on to instruction fine-tuning and finally DPO. If that works well, you may want to explore other models or finetuning strategies.

Part B: Language models in practice

(30 %)

Deliverable 3 – Multilingual Model

Use a multilingual model and prompt translations for one of your target languages.

Notebooks: *LaD3..._instructions.ipynb* and *LaD3..._multilingual.ipynb*

- **Option 1: Qualitative Evaluation**

Select 40 prompts such that you **systematically** vary one characteristic (e.g., register, domain, syntactic complexity) and analyze how it affects the translation quality. For example, you could design minimal pairs of sentences that only differ in tense or gender.

- **Option 2: Quantitative Analysis**

Extract translations from a parallel corpus (e.g., from the **Workshop of Machine translation**) and evaluate the quality using the BLEU score. Reflect on the strengths and weaknesses of the BLEU score.

Part C: Reflection

(5 %)

Deliverable 4

Provide a brief summary on how you approached the problem as a group. What challenges did you encounter? What surprised you? What did you find most interesting?